

Counting Principles (7.5)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 12

_____ How many 3-letter codes are possible if each position is any letter A–Z?

ANSWER: 30

_____ Evaluate $\binom{12}{4}$.

ANSWER: 60

_____ Evaluate $\binom{10}{2}$.

ANSWER: 336

_____ Evaluate $\binom{5}{5}$.

ANSWER: 17,576 # _____ Evaluate ${}_5P_3$.	ANSWER: 420 # _____ Evaluate ${}_8P_3$.
ANSWER: 24 # _____ Evaluate ${}_6P_2$.	ANSWER: 1 # _____ An outfit uses one of 3 shirts and one of 4 pairs of pants. How many outfits are possible?
ANSWER: 45 # _____ Evaluate $4!$.	ANSWER: 495 # _____ How many distinguishable arrangements of the letters in TATTOOS are there?